FLAIR Task Schematic

Extubation Failure within 24 Hours (Task T3)

FLAIR Benchmark

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Abstract

We specify the FLAIR extubation-failure task: at the moment of the first confirmed extubation, predict whether the patient will accumulate less than one ventilator-free day in the next 24 hours — i.e., die or get reintubated. This is a one-per-stay task anchored at a single per-hospitalization event. The interesting algorithmic ingredient is the consensus-window IMV episode detector (`_ventilation_episodes.detect_imv_episodes`), which distinguishes a true extubation from sensor flicker, tracheostomy sentinels, and intermittent ventilator swaps. Ambiguous censoring (discharged alive less than 24h post-extubation with no observed event) is dropped to avoid positive-unlabeled bias (Epstein et al., 1997; Thille et al., 2011).

1 Consensus-window IMV episode detection

A CLIF `respiratory_support` row is a device-status sample, not a clinical event. A single misclassified row (brief sensor disconnect, vent swap, transient downgrade) should not be counted as intubation or extubation. `detect_imv_episodes` confirms each transition with a forward consensus window:

IMV start.
A device `imv` row at time t opens an episode iff no contradicting non-IMV row falls in $(t, t + \texttt{INTUB_WINDOW_MIN}]$ with $\texttt{INTUB_WINDOW_MIN} = 15$ minutes.

Episode close (true extubation).
A non-IMV row at time t (while an episode is open) closes that episode iff no contradicting IMV row falls in $(t, t + \texttt{EXTUB_WINDOW_MIN}]$ with $\texttt{EXTUB_WINDOW_MIN} = 30$ minutes.

Trach sentinel.
A row with `device_category` = `"trach collar"` or `tracheostomy` = `true` closes any open episode at its timestamp and halts further counting for the hospitalization. If the trach sentinel fires before any IMV row, `is_pre_existing_trach` = `true`.

The detector emits one row per confirmed episode: `(hospitalization_id, episode_index, episode_start_dttm, episode_end_dttm, is_pre_existing_trach)`. The `episode_end_dttm` field is null iff the patient is still on IMV at the end of record or the episode was closed by a trach sentinel that did not first pass through a non-IMV transition.

2 FLAIR operational definition

2.1 Index time and label

Let H_i denote a hospitalization that survives the cohort filters (§2.2). Let $E_i^1 = (s_i^1, e_i^1)$ be its first IMV episode and $E_i^2 = (s_i^2, e_i^2)$ its second (if any).

Prediction time (t^i_{extub}).
$t^i_{\text{extub}} = e_i^1 =$ `episode_end_dttm` of the first episode. Exactly one decision point per hospitalization (*one-per-stay*).

Reintubation time.
$t^i_{\text{reint}} = s_i^2$ if E_i^2 exists, $+\infty$ otherwise.

Death time.
t^i_{death} from `hospitalization.death_dttm`; $+\infty$ if null.

Label horizon.
$(t^i_{\text{extub}},\, t^i_{\text{extub}} + 24\text{h}]$.

Label (`label_extubation_failure_24h`).

$$y_i \;=\; \mathbb{1}\Big[t^i_{\text{death}} \in (t^i_{\text{extub}},\, t^i_{\text{extub}} + 24\text{h}\,] \;\;\vee\;\; t^i_{\text{reint}} \in (t^i_{\text{extub}},\, t^i_{\text{extub}} + 24\text{h}\,]\Big]\,.$$

Observability end.
$t^i_{\text{obs_end}} = \texttt{coalesce}(t^i_{\text{death}},\, t_i^{\text{disch}})$.

Censoring drop.
Keep row iff

$$t^i_{\text{death}} \in (t^i_{\text{extub}},\, t^i_{\text{extub}} + 24\text{h}\,] \;\;\vee\;\; t^i_{\text{reint}} \in (t^i_{\text{extub}},\, t^i_{\text{extub}} + 24\text{h}\,] \;\;\vee\;\; t^i_{\text{obs_end}} \;\geq\; t^i_{\text{extub}} + 24\text{h}.$$

A patient discharged alive less than 24 hours after extubation with no observed death or reintubation has *unobserved* outcome; treating that as $y_i = 0$ would inject positive-unlabeled bias.

2.2 Cohort

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- ≥ 1 ICU ADT record.
- First IMV episode (`episode_index` $= 1$) has non-null `episode_end_dttm` (i.e., a confirmed extubation exists per §1).
- **Exclude** pre-existing tracheostomy (`is_pre_existing_trach`).
- **Exclude** first episode closed by a trach sentinel ($\pm$1min tolerance via `TRACH_CLOSURE_TOLERANCE_MIN` $= 1$; `_drop_trach_closed` in `task3_extubation_failure_24h.py`).
- **Exclude** ambiguous censoring (discharged alive < 24h post-extubation with no death and no reintubation).

2.3 CLIF tables and categories used

Task T3 reads four CLIF v2.1 tables. The cohort entry filter is on `adt.location_category`; the index event itself is derived from two category-column predicates in `respiratory_support`.

CLIF table	Category column	Literal values kept (case-insensitive)
`hospitalization`	—	cohort/demographics join; no category filter
`patient`	—	demographics join; no category filter
`adt`	`location_category`	{`"icu"`}
`respiratory_support`	`device_category`	{`"imv"`} classifies a row as IMV; any other value is non-IMV;
	trach sentinel: `device_category` = `"trach collar"` ∨ `tracheostomy` = `true`†	

Table 1: CLIF surface for Task T3. Both the IMV-vs-non-IMV classifier and the trach sentinel live in `respiratory_support`; the consensus-window logic of §1 runs on top of these two predicates.

† `tracheostomy` is a *boolean* column, not a category. The trach sentinel fires when either the categorical `device_category` equals `"trach collar"` OR the boolean `tracheostomy` is `true`. A site whose CLIF `respiratory_support` export omits one of these signals can still detect tracheostomies via the other; sites that omit both will silently include trach-decannulation events in the extubation cohort.

Columns selected from each table:

- `hospitalization`: `hospitalization_id`, `patient_id`, `admission_dttm`, `discharge_dttm`, `age_at_admission`, `discharge_category` (loaded but not filtered).
- `patient`: `patient_id`, `sex_category`, `race_category`, `ethnicity_category`, `death_dttm`.
- `adt`: `hospitalization_id`, `location_category`, `in_dttm`, `out_dttm`.
- `respiratory_support`: `hospitalization_id`, `recorded_dttm`, `device_category`, `tracheostomy`.

3 Algorithmic flow

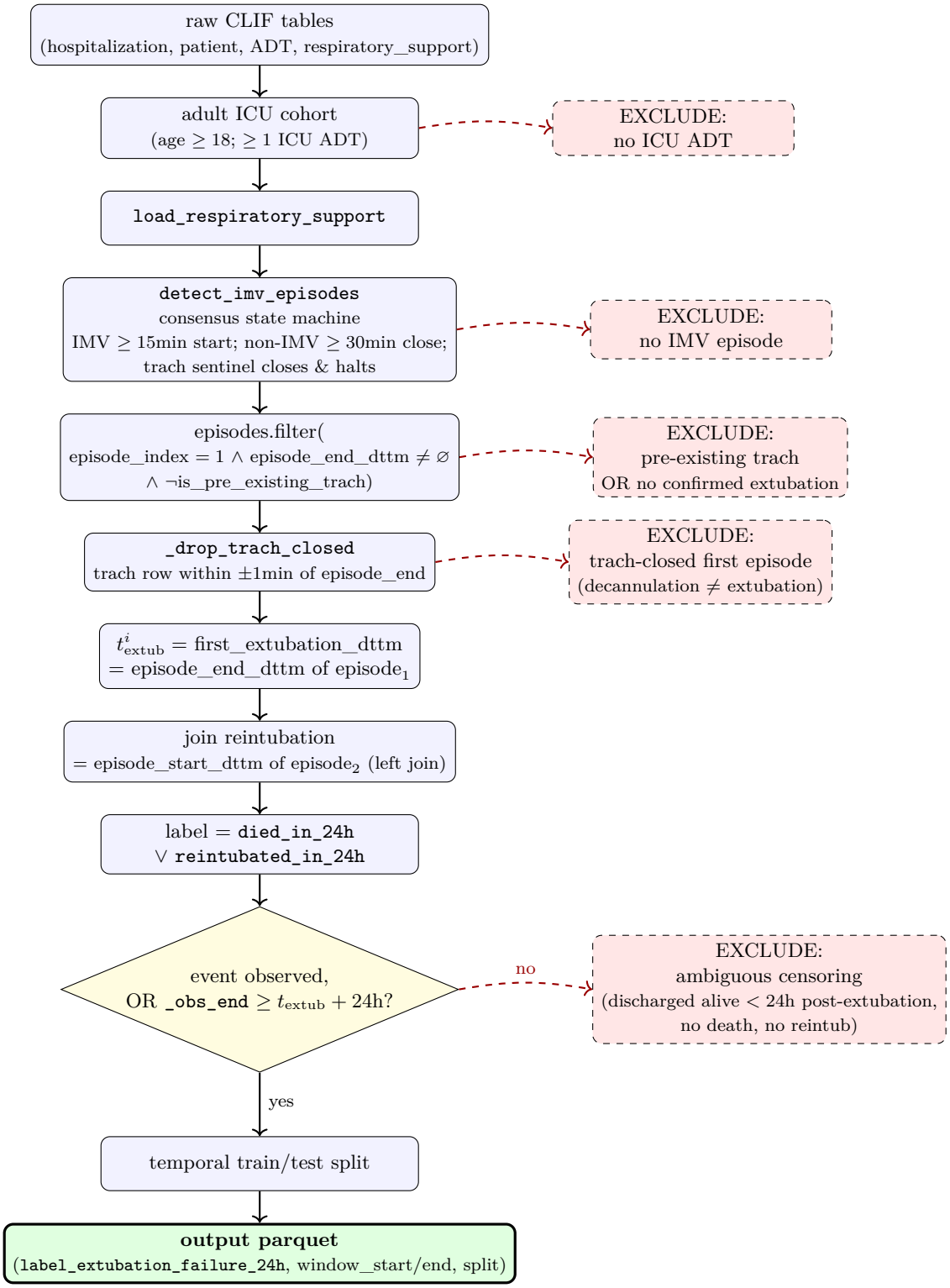


Figure 1: Build pipeline for Task T3. Solid arrows: main flow. Dashed red arrows: exclusions. The diamond enforces the censoring rule: any hospitalization whose observation window closes before $t^i_{\text{extub}} + 24\text{h}$ without an observed death or reintubation is dropped, since its label is unobservable.

Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt`, `respiratory_support`.
2. Apply adult-ICU cohort filters.
3. `load_respiratory_support(cfg, hosp_ids)` returns all device-status rows for the cohort.
4. Run `detect_imv_episodes(resp)` (§1).
5. Filter to first episodes that ended in a confirmed extubation: `episode_index` $= 1 \wedge$ `episode_end_dttm` $\neq \varnothing \ \wedge \ \neg$`is_pre_existing_trach`.
6. Drop trach-closed first episodes via `_drop_trach_closed`: any first episode whose `episode_end_dttm` coincides with a `trach collar` or `tracheostomy=true` row within ± 1 minute is removed (decannulation $\neq$ extubation).
7. Extract second-episode starts as t^i_{reint} = `episode_start_dttm` where `episode_index` $= 2$ (left join; null if none).
8. Join hospitalization-level t^i_{death} and t^{disch}_i.
9. Compute $y_i =$ `_died` $\vee$ `_reintub` per the formulas above.
10. Drop ambiguous censoring rows (the diamond in Figure 1).
11. Set `window_start` $= t^i_{\text{extub}} - 24$h, `window_end` $= t^i_{\text{extub}}$ (fixed 24h lookback).
12. Apply temporal train/test split via `flair_benchmark._split.assign_split`, keyed on t^i_{extub}.
13. Select `OUTPUT_COLS` and persist as parquet.

4 Design choices and rationale

Why 15-minute / 30-minute consensus windows. RespiratorySupport in CLIF is a device-status sample: brief sensor dropouts, vent swaps, and transient downgrades produce single rows that look like state transitions but are not. The 15-minute IMV consensus on the start side and the 30-minute non-IMV consensus on the close side filter these artifacts while still resolving real extubations within a clinically reasonable margin. The asymmetric choice (15 in, 30 out) reflects the asymmetric cost: a missed intubation start would lose only a small slice of input-feature window, whereas a false extubation close anchors the entire label.

Why drop trach-closed first episodes. The state machine treats a trach sentinel as an episode close so the counting can halt cleanly, but a tracheostomy decannulation is not the clinical event this task predicts. Modeling it as an extubation would mix a planned airway management procedure with the unplanned reintubation-vs-failure decision and inflate the cohort with patients whose 24h outcome is dominated by trach-specific physiology rather than spontaneous-breathing readiness.

Why drop ambiguous censoring. A patient extubated at 03:00 and discharged at 18:00 the same day with no death and no reintubation has only 15 hours of post-extubation observability. Labeling them $y_i = 0$ would treat the absence of observed failure as evidence of success when it is just censoring. This is a textbook positive-unlabeled trap; the cohort filter resolves it by dropping the unobservable row entirely. The cost is a small positive-skew in the retained cohort (patients with longer post-extubation observation are slightly sicker on average), which the main paper's Discussion acknowledges.

Why one-per-stay (not continuous). The decision to extubate is itself the clinical handle. Predictions at arbitrary timesteps before the actual extubation are not actionable because they would be evaluated against a different intervention (extubation at a different time). Restricting to the actual extubation moment makes the label well-defined and the comparison across hospitalizations meaningful.

Prevalence implications. MIMIC-CLIF: 26,771 extubation episodes pass the cohort filters, 1,456 positive (5.4%; see `docs/paper/outline.tex` Table 1). Reporting AUPRC alongside AUROC is mandatory.

5 Out of scope

- Second-, third-, and later-extubation decision points.
- Non-invasive-ventilation (NIV) as a failure mode (currently only death and reintubation count).
- A continuous variant that predicts extubation failure ahead of the actual extubation.
- Curated drug-list or ventilator-mode feature design.

References

Scott K. Epstein, Mary L. Nevins, and Jean Chung. Effect of failed extubation on the outcome of mechanical ventilation. *Chest*, 112(1):186–192, 1997. doi: 10.1378/chest.112.1.186.

Arnaud W. Thille, Anatole Harrois, Frédérique Schortgen, Christian Brun-Buisson, and Laurent Brochard. Outcomes of extubation failure in medical intensive care unit patients. *Critical Care Medicine*, 39(12):2612–2618, 2011. doi: 10.1097/CCM.0b013e3182282a5a.